

Comprehensive End-User Documentation & Operating Manual

Step-by-Step Operating Guide for Tenants, Appliance Providers, Logistics Technicians & Admins

Academic and strategic research artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2). Demonstrating circular economy transition, machine learning churn mitigation, FunkSVD recommendation, and localized generative AI edge deployment.

Document Ref:	USER-SDC2-2026-V2.4	Publication:	September 2026
Project Title:	RentAI / SDC2 Autonomous Rental Platform	Peer Review:	Academic Capstone & IEEE/ACM Format
Principal Authors:	Advanced AI & Systems Research Group	Affiliation:	Department of Computer Science & Engineering
Classification:	Scholarly Research & Field Engineering	Status:	PUBLISHED & ARCHIVED

1. Tenant User Guide

1.1 Account Registration & KYC Verification

Tenants register using their email address and secure password. To initiate an appliance booking, tenants submit digital KYC documentation (Government ID verification), ensuring compliance and automated security deposit escrow protection.

1.2 Exploring Inventory & Category Filtering

The catalog interface allows multi-criteria filtering:

- **Category Selection:** ACs, Refrigerators, Washing Machines, Televisions, Microwaves, and Combos.
- **Price Ceiling Slider:** Sets maximum daily or monthly subscription limits.
- **Availability Toggle:** Immediately filters for units in warehouse stock for immediate dispatch.

1.3 Interactive AI Assistant (Rentova AI)

Click the floating violet assistant icon in the bottom-right corner. Tenants can ask natural language queries such as:

"Show me double-door refrigerators under ₹300 per day with free delivery in Hyderabad."

The assistant responds with real inventory listings, transparent daily/monthly rental rates, and deposit requirements.

2. Appliance Provider / Host Guide

Appliance owners list durable inventory on the marketplace. Hosts define monthly rental yields, specify maintenance schedules, and monitor recurring lease disbursements via the Host Earnings Dashboard.

3. Logistics & Field Engineering Guide

Field technicians manage scheduled deliveries, install grounding circuits, conduct digital pre-delivery physical condition inspections, and document client handover signatures directly within the dispatch interface.

4. Platform Administrator Guide

Administrators access platform-wide operational telemetry:

- **Churn Risk Monitor:** Identifies tenants with churn probabilities $\geq 65\%$ and triggers tailored promotional incentives.
- **Fleet Logistics Dispatch:** Tracks units across transit, active lease, and return inspection phases.
- **Ollama AI Supervisor:** Displays active model status (``rentai-llm:latest``, ``qwen2.5:0.5b``) with one-click background restart capabilities.